

Within the flat 6D Conformal Spacetime Algebra system ($Cl(4, 1, 1)$), the release of light by a Light-Emitting Diode (LED) is modeled without the need for second-quantized wave-particle operators. Instead, it is treated as a classical, first-order dynamical phase transition of an electron soliton settling into a localized potential well—a process governed by the transient violation and subsequent restoration of the **Radiationless Condition** [5.1].

This physical process is detailed below using convective wavefront trains and encounter-timing mechanics at the boundary of a receiving system.

Section 5.5: Light-Emitting Diodes (LED) & Convective Wavefront Emission

An LED releases light when a mobile conduction electron soliton is captured by a localized positive vacancy (“hole”) in a semiconductor lattice, dropping into a bound valence state. This transition is governed by the transient violation and subsequent restoration of the **Radiationless Condition** [5.1].

5.5.1 The Lattice and “Hole” Potential Well In a semiconductor crystal lattice, the atoms form a periodic, phase-locked network of positive tetrahedral cores [5.4] and outer valence electron loops. * A **hole** is modeled as a localized positive electrostatic potential vacancy $V_{\text{hole}}(r)$ in the valence band. * This vacancy produces an effective static electric field $E_{\text{hole}}(r)$ and a localized magnetic gradient $B_{\text{hole}}(r)$ generated by the orbital rotation of the surrounding lattice electrons.

5.5.2 Conduction to Valence Energy Transition A conduction electron is a mobile, self-bound current loop of radius r_e [4.1] moving through the lattice with a drift velocity u_{drift} . Its total energy in the conduction state is:

$$\mathcal{H}_{\text{conduction}} = K_{\text{drift}} + \mathcal{H}_{\text{self}}$$

When the electron loop is captured by the positive potential well of the hole, it transitions into a bound valence state of lower energy:

$$\mathcal{H}_{\text{valence}} = \mathcal{H}_{\text{bound}}$$

The energy difference between these two stable states is the semiconductor bandgap energy E_g :

$$E_g = \mathcal{H}_{\text{conduction}} - \mathcal{H}_{\text{valence}}$$

5.5.3 Transient Violation of the Radiationless Condition A stable, bound electron loop does not radiate because its orbital frequency ω_{orbit} matches its spin precession frequency ω_{prec} , enabling phase-cancellation of its outer fields [5.1].

However, during the transition interval Δt_{trans} as the electron loop falls into the hole, this balance is temporarily disrupted: 1. **Phase Mismatch:** The electron's orbital frequency accelerates as it sinks into the potential well, while its spin plane precession lags behind the local magnetic gradient B_{hole} :

$$\omega_{\text{orbit}}(t) \neq \omega_{\text{prec}}(t)$$

2. **Transient Current Source (J_{trans}):** This phase mismatch breaks the *Radiationless Condition*. The phase-cancellation of the loop's outer fields fails, generating an asymmetric, periodically oscillating transient current density J_{trans} :

$$J_{\text{trans}}(t) = J_0 \cos(\Delta\omega \cdot t) e^{-\frac{t^2}{2\tau^2}}$$

where: * $\Delta\omega = \omega_{\text{orbit}} - \omega_{\text{prec}}$ is the transition frequency of the radiating loop. * $\tau \approx \Delta t_{\text{trans}}$ is the transition timescale (the decay envelope of the emission).

This oscillating current acts as a micro-antenna, emitting a series of discrete, concentric bivector wavefronts.

5.5.4 Emission of the Convective Wavefront Train The transient current J_{trans} acts as the source term in the master field equation [2.4]:

$$DF_{\text{wave}} = J_{\text{trans}}$$

Since the gravitational coupling α is negligible at these atomic energy scales, the emitted field bivector is purely electromagnetic:

$$F_{\text{wave}} = E_{\text{wave}}e_4 + cB_{\text{wave}}I_3$$

Because the source J_{trans} is active only during the transition interval τ , the emission terminates once the electron settles and the radiationless condition is restored. This results in a finite train of N discrete wavefronts.

In vacuum, this wavefront train propagates ballistically relative to the semiconductor crystal lattice at a velocity of $c \pm u_{\text{lattice}}$:

$$DF_{\text{wave}} = 0 \implies \nabla \times E_{\text{wave}} = -\frac{\partial B_{\text{wave}}}{\partial t} - (u_{\text{lattice}} \cdot \nabla)B_{\text{wave}}$$

5.5.5 Encounter-Based Frequency Quantization Because the electron soliton's structural geometry [4.1] can only exist in stable, non-radiating states (which are geometrically quantized by the boundary conditions of the periodic crystal lattice), the transition must occur as a single, discrete step. The transition frequency $\Delta\omega$ determines the rate at which successive wavefronts are generated by the source.

The period of oscillation of the emitting source is:

$$T_{\text{emit}} = \frac{2\pi}{\Delta\omega}$$

The spatial distance λ (wavelength) between successive wavefronts in the propagating train is determined by the emission period and the ballistic speed relative to the lattice:

$$\lambda = (c \pm u_{\text{lattice}})T_{\text{emit}} = \frac{2\pi(c \pm u_{\text{lattice}})}{\Delta\omega}$$

A receiving system (such as an absorbing lattice or detector) moving at velocity v_{receiver} intersects this propagating train of wavefronts. The frequency ν registered by the receiving system is defined strictly by the **timing of encounters** with these successive wavefronts at the receiver's boundary:

$$\nu = \frac{1}{\Delta t_{\text{encounter}}} = \frac{c \pm u_{\text{lattice}} \mp v_{\text{receiver}}}{\lambda}$$

Substituting λ into the encounter rate equation:

$$\nu = \frac{\Delta\omega}{2\pi} \left(\frac{c \pm u_{\text{lattice}} \mp v_{\text{receiver}}}{c \pm u_{\text{lattice}}} \right)$$

This relation expresses the registered frequency as an encounter rate with physical wavefronts, incorporating the ballistic Doppler shift under classical acoustic-like kinematics.

The total energy of the convective wavefront train is tied to the discrete energy step E_g of the transition:

$$E_g = h \left(\frac{\Delta\omega}{2\pi} \right)$$